

Alex Xu

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Education

University of Waterloo

2023 – 2028

Computer Engineering, Honours, Co-operative Program

Skills

Languages: JavaScript, TypeScript, Python, Java, C++

Web: Docker, Socket.IO, Tailwind, Bootstrap

Fullstack: React, NextJS, Django, Flask, Firebase, Nginx

Game Dev: Unity, Three.js, OpenGL

Cloud & DBs: GCP, AWS, MongoDB, PostgreSQL

Developer Tools: IntelliJ IDEA, PyCharm, VS Code, GitHub

Experience

Apollo Intelligence Inc

Sept 2024 – Dec 2024

Fullstack Developer & Designer

NextJS, Supabase, React, Node, Python, NextUI, MUI, RESTful

Apollo Agents

- Reached the **Y Combinator interview stage**, an achievement with a success rate of **less than 1%**.
- Built a chat interface using **React.js** and **Blockly**, hosted services over **Python**, and **Supabase**.
- Designed a **recording-based workflow system** that translates user's recorded browser actions to **VPL**.
- Integrated and **fine-tuned** models on **OpenAI & Claude's API**, **reducing token output by 50%** with higher stability.

Apollo Learn (<https://www.apollolearn.ca/>)

- Developed the frontend of an **AI-powered** education platform using **NextJS** and **TS**, integrating **NextUI** and **MUI**.
- Implemented assignment submission, grading, feedback functionalities with **Supabase** & **AWS** for backend.
- Created an **AI-grading system** that analyzes rubric and provides feedback to student automatically, saving 70% time.
- Finished the entire workflow of an **Auth System**, including but not limited to **Sign Up** and **Password Reset**.
- Developed a **replay system** to track assignment edits for anti-plagiarism detection and originality verification.
- Deployed the platform for real-world use by a Germany class in **Queen's University** during **Sept-Dec 2024**.

Projects

Minecraft Remake in Java

Oct 2021 – Jan 2022

https://github.com/AlexZihaoXu/Minecraft_AE

Java, LWJGL, GLFW, OpenGL, GLSL

- Developed a complete remake of **Minecraft** from scratch with **Java** & **GLFW**. **ALL ASPECTS** of the project were **self-taught** using resources like LearnOpenGL. No external code was copied from YouTube or other GitHub repositories.
- Built a custom rendering engine with **OpenGL** and **GLSL** shaders on **LWJGL**.
- Implemented **terrain generation** using **Perlin noise**, optimized with **multi-threading** for improved performance.
- Added a **lighting system** using **Breadth-First Search (BFS)**.

Minecraft Remake in C++

Oct 2022 – Oct 2023

<https://github.com/AlexZihaoXu/minecraft.cpp>

CMake, C++, GLSL, STB, OpenGL, GLFW

- Built the entire rendering engine with **OpenGL** and **GLSL** shaders on the **GLFW** library.
- Implemented **meshing system** that only adds visible faces to the geometry, greatly **reduces VRAM usage**.
- Tested using **Vanguard**, ensuring **no memory leaks** during runtime.

Ice Knight (Game)

Feb 2025

<https://me.alex-xu.site/unity/ice-knight/>

Unity, C#

- Collaborated **with 2 other developers** during UWaterloo Game Dev Club W2025 Game Jam in 3 days.
- Implemented multiple core game features including but not limited to **player control**, **enemy AI**, and **projectiles**.
- Created and worked on various VFX such as **camera shake**, **particles**, **shadow casting**, **2D lighting** and some post-processing effect such as **glowing**. **Drew** and animated ALL entities in the game.

Party Physics (Game) (Grade 12 Computer Science Final Project)

Oct 2022 – Jan 2023

<https://github.com/AlexZihaoXu/PartyPhysics>

Java, Java2D, Dyn4J, Socket, TCP

- Developed a **multiplayer game** similar to **Stick Fight** from scratch, fully *cross-platform*.
- Engineered a **custom active rendering framework** in **Java Swing** with **anti-aliasing**, **particle system**, and **interpolated animations** for smooth physics at a lower tick rate.
- Built multiplayer functionality on a **custom TCP networking framework** with an abstraction layer for efficient **serialization** and **deserialization**, supporting **300+ moving objects** with **sub-10 ms latency**.
- Integrated real-time physics simulation with the **Dyn4J** library, minimizing network latency conflicts.

SoulKnight Clone

Feb 2020 – Mar 2020

Personal Project

Python, Pygame

- Created a tool to **edit animations** and **adjust frame offsets** for extracted sprites.
- Developed a **SoulKnight remake** in **Pygame**, optimizing rendering with a **caching system** for **5x more framerate**.
- Designed and integrated **randomly generated dungeons** and **enemy AI**.

FPV Drone Project

Oct 2024 – Nov 2024

Personal Project

DJI O3 Air Unit, ExpressLRS, FPV Hardware

- Built an **FPV drone** from scratch, integrated **DJI O3 Air Unit** for high-quality video transmission with **30 ms latency**.
- Configured the drone with **ExpressLRS** as the TX/RX system for reliable long-range control.
- Used **lower KV motors** and a **compact frame** to improve efficiency, achieving an **8-minute flight time**.
- Installed a **GPS module** with **auto-return-to-home (RTH)** for navigation and recovery during signal loss.
- **Designed** the drone for **0–60 km/h acceleration in under 1 second** with a **4 km+ range**, limited by battery capacity.

Robotic Arm Project

Sept 2023 – Nov 2023

University of Waterloo – ECE 1A – ECE 198

- Developed a robotic arm from scratch for the **ECE 198** project using the **STM32 Nucleo** board, programming the system and UI in **C** for real-time control. **Achieved a final course grade of 94%**.
- Designed and refined the robotic arm using **Fusion 360**, optimizing the model through **5 iterations**.
- Created a custom gear set to enhance torque transfer, improving **stepper motor** efficiency. Designed a box-shaped control unit with an integrated wind tunnel for **improved cooling**.
- Implemented an **asynchronous I2C LCD driver** and configured an **ADC DMA joystick**, enhancing system responsiveness and preventing process blocking.
- Utilized **spring physics to smooth** arm movement, reducing vibration and instability during operation.
- Implemented a **reverse-kinematics** system, abstracting a **coordinate system** to achieve easier control.

D2L Backup Utilities

Mar 2023 – Apr 2023

Personal Project

Python, Django, JavaScript

- Leveraged **Django's ORM** to model the course structure on **D2L**, utilizing **Django's Admin features** to manage and filter course content effectively, including sections and files.
- Developed a **GraphQL API** using graphene-django to enable advanced and flexible data queries and mutations.
- Designed a **JavaScript** program that autonomously analyzes the structure of a D2L course from the Content page, outputting the structure with downloadable links in a nested JSON format.
- Created a **Python script** that utilizes nested **JSON** data generated by the JavaScript tool to download and populate all course data into the **database** via **Django's ORM**.